

X-Ray Radiation: Bragg Diffraction, Absorption and Moseley's Law

Áslaug Edda Kristjánsdóttir, Fríða Margrét Guðmundsdóttir og Hildur Björk Búadóttir¹

¹*Raunvísindadeild, Háskóli Íslands*

(Dated: March 23, 2026)

This experiment used a molybdenum X-ray tube and a NaCl monocrystal to study Bragg diffraction, X-ray absorption, and Moseley's law. The characteristic wavelengths were determined as $\lambda_\alpha = 71.67$ pm and $\lambda_\beta = 63.85$ pm. Due to instrumental broadening ($\delta\theta \approx 0.391^\circ$), the K_α doublet could not be resolved, with calculations showing that an order of $n \geq 6$ would be required for separation. The zirconium K-edge was found at 68.74 pm, and the absorption cross-section was shown to follow $\tau_a \propto \lambda^{2.5}$ and $\tau_a \propto Z^{3.37}$. A Moseley plot of $\sqrt{E_K}$ against Z for several metals yielded an experimental Rydberg constant of $R_{\text{calc}} = 1.104 \times 10^7 \text{ m}^{-1}$ and a screening constant of $\sigma_K \approx 3.71$. These results confirm the predicted relationship between atomic number and inner-shell binding energies, with the experimental Rydberg constant agreeing with the theoretical value to within 0.6%.

INTRODUCTION

The discovery of X-rays by Wilhelm Conrad Röntgen in 1895 marked a major breakthrough in atomic physics and allowed scientists to investigate the structure of matter in detail. Since then, X-rays have played a crucial role in determining atomic, molecular and crystal structures, as their wavelength is comparable to atomic distances [1].

A key feature of X-rays is their interaction with matter through diffraction and absorption. Characteristic X-ray lines are produced when inner-shell vacancies in atoms are filled by electrons from higher energy levels, while absorption edges occur when the photon energy becomes sufficient to ionize inner-shell electrons. These processes provide direct information about atomic energy levels and binding energies [1].

In this experiment, Bragg diffraction from a NaCl crystal was used to measure the characteristic spectrum of a molybdenum X-ray tube and determine the wavelengths and photon energies of the K_α and K_β lines. Absorption measurements in metal foils were then used to identify absorption edges and examine the expected wavelength and atomic-number dependence of X-ray attenuation. Finally, Moseley's law was tested using measured K-edge data in order to determine an experimental value of the Rydberg constant.

EXPERIMENTAL APPARATUS AND METHOD

The experiment was performed using a Leybold X-ray apparatus equipped with a molybdenum anode and a NaCl monocrystal mounted on the built-in goniometer. The measurements were carried out in Bragg configuration with 2θ coupling between the crystal and detector. The X-ray tube was operated at an acceleration voltage of $U = 35$ kV and an emission current of $I = 1.00$ mA throughout the experiment. The counting time was 10 s per measurement point. For NaCl, the lattice constant is $a_0 = 564.02$ pm, corresponding to a lattice plane spacing

of $d = a_0/2 = 282.01$ pm.

Bragg diffraction spectrum

The diffraction spectrum of the Mo anode was recorded using the NaCl crystal. The Bragg angle θ was scanned in steps of 0.1° over an angular range of approximately 5.6° to 30.5° , sufficient to observe the K_α and K_β lines up to the fourth diffraction order. The wavelength was determined from Bragg's law

$$n\lambda = 2d \sin \theta, \quad (1)$$

and the corresponding photon energies were calculated using

$$E = \frac{hc}{\lambda}. \quad (2)$$

To estimate whether the $K_{\alpha 1}$ and $K_{\alpha 2}$ doublet could be resolved, the angular separation was calculated from

$$\Delta\theta = \frac{n\Delta\lambda}{2d \cos \theta}. \quad (3)$$

To physically resolve this fine structure, the theoretical angular separation $\Delta\theta$ must exceed the intrinsic instrumental peak width $\delta\theta$. The calculated angular separation was therefore compared with the measured peak width in order to assess whether the doublet could be resolved.

Wavelength dependence of absorption

The transmittance of copper and zirconium foils was measured as a function of wavelength. The angle was scanned from 4.2° to 12.5° in steps of 0.1° . First, a reference spectrum without absorber, I_0 , was recorded. The measurement was then repeated with a Cu foil, giving I_{Cu} , and with a Zr foil, giving I_{Zr} . The attenuation follows Lambert–Beer's law

$$I = I_0 e^{-\mu x}, \quad (4)$$

where μ is the linear attenuation coefficient and x is the foil thickness. The total attenuation coefficient is given by

$$\mu = \tau + \sigma, \quad (5)$$

where τ is the absorption coefficient and σ the scattering coefficient. Outside the absorption edges, the theoretical model predicts that the absorption cross-section follows

$$\tau_a = C\lambda^3 Z^4. \quad (6)$$

The measured spectra were analyzed using the quantities I_0 , I_x , and $\ln(I_x/I_0)$ as functions of wavelength in order to examine the wavelength dependence of the absorption outside the absorption edges.

Atomic number dependence (Z^4 law)

To investigate the dependence of absorption on atomic number, measurements were performed for Al, Fe, Cu, Zr, and Ag foils at a fixed glancing angle of $\theta = 4.1^\circ$, corresponding to a constant wavelength of approximately $\lambda \approx 40.3$ pm according to Bragg's law. This choice allows the effect of the atomic number Z to be examined at fixed wavelength, as required by the theoretical model in Eq. (6).

From the measured transmission, the linear attenuation coefficient μ was obtained using Lambert–Beer's law. The atomic absorption cross-section was then calculated as

$$\tau_a = \frac{\mu}{\rho} \frac{A}{N_A} - \sigma_a \quad (7)$$

where ρ is the density, A is the atomic mass, N_A is Avogadro's constant, and the scattering contribution was approximated as

$$\sigma_a \approx 0.2 \cdot \frac{A}{N_A} \text{ cm}^2$$

[2]. A log–log plot of τ_a versus Z was then used to test the expected Z^4 dependence.

Moseley's law

To test Moseley's law, the K-absorption edges of Zr, Mo, Ag, and In were measured. In addition, Fe was included in the combined intensity spectrum for comparison, but was not used in the Moseley analysis since its K-edge lies outside the scanned wavelength range. The Bragg angle was scanned from 3.7° to 7.5° in steps of 0.1° . Combined intensity spectra and the corresponding transmission spectra were then used to identify the K-edge positions.

The corresponding edge wavelengths λ_K were obtained using Bragg's law (Eq. (1)). For the Moseley analysis, these wavelengths were converted to K-edge energies using Eq. (2) with $\lambda = \lambda_K$. Moseley's law relates the K-edge wavelength to the atomic number as

$$\sqrt{\frac{1}{\lambda_K}} = \sqrt{R_\infty}(Z - \sigma_K) \quad (8)$$

where R_∞ is the Rydberg constant and σ_K is the screening constant of the K-shell. Using $E_K = hc/\lambda_K$, this may equivalently be written as

$$\sqrt{E_K} = \sqrt{hcR_\infty}(Z - \sigma_K) \quad (9)$$

A linear regression of the Moseley plot was then used to determine the experimental value of the Rydberg constant and the screening constant.

RESULTS AND DISCUSSION

Bragg diffraction spectrum

Figure 1 displays the measured X-ray diffraction spectrum and the characteristic molybdenum peaks for diffraction orders up to $n = 4$. Using Equation (1), the characteristic wavelengths were determined to be $\lambda_\beta = 63.85$ pm and $\lambda_\alpha = 71.67$ pm. These values were derived from the first-order diffraction peaks ($n = 1$). This order provides the highest intensity and yields a single, well-defined maximum before fine-structure broadening complicates the peak shape at higher orders. These results align closely with the specifications of the literature $\lambda_\alpha = 71.08$ pm and $\lambda_\beta = 63.09$ pm [3], with an estimated measurement precision below 1 pm. Furthermore, by applying Equation (2), the corresponding photon energies were calculated as $E_\beta = 19.42$ keV and $E_\alpha = 17.30$ keV.

The K_α emission is a doublet consisting of $K_{\alpha 1}$ and $K_{\alpha 2}$ lines with a wavelength difference of $\Delta\lambda = 0.43$ pm [4]. By combining Equation (1) and Equation (3) to eliminate the Bragg angle θ , a direct condition for the minimum required diffraction order n is derived:

$$n \geq \frac{2d \cdot \delta\theta}{\sqrt{\Delta\lambda^2 + (\lambda \cdot \delta\theta)^2}} \quad (10)$$

From the measured first-order spectrum, the instrumental broadening caused by collimator slit width and beam divergence was estimated as $\delta\theta \approx 0.391^\circ$. Evaluating Equation (10) with this angular width yields a theoretical minimum order of $n \geq 5.94$. This demonstrates why the doublet remains completely unresolved in the first four diffraction orders. Since the diffraction order n must be an integer, a diffraction order of $n \geq 6$ is required for complete separation in our apparatus, although literature indicates that partial resolution becomes experimentally observable at $n = 5$ [5].

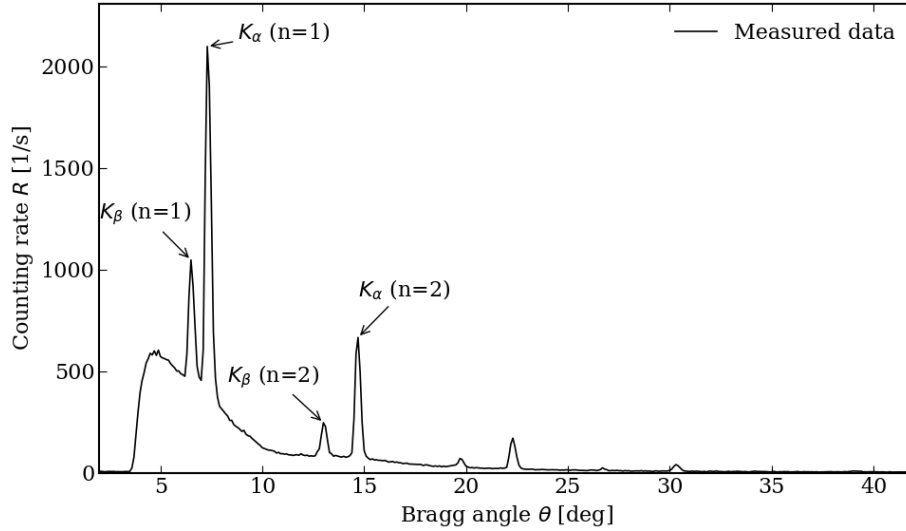


Figure 1: X-ray Bragg diffraction spectrum from a NaCl crystal, showing the counting rate R versus the Bragg angle θ . The characteristic K_α and K_β lines of the molybdenum anode are clearly visible up to the fourth diffraction order.

Wavelength dependence of absorption

Figure 2 displays the measured X-ray intensity as a function of wavelength for the unattenuated beam (I_0) and beams attenuated by copper (I_{Cu}) and zirconium (I_{Zr}) foils. The transmission spectra clearly demonstrate how the intensity decreases as it passes through the metal foils due to photoelectric absorption and scattering processes [6].

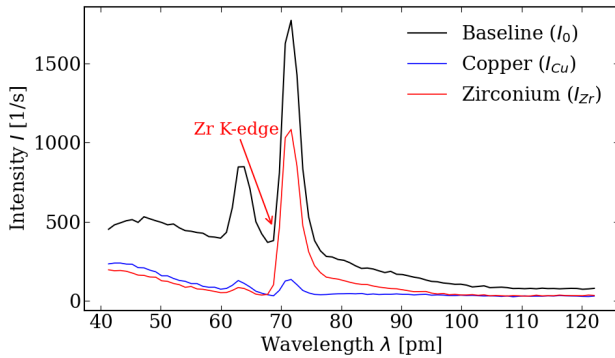


Figure 2: X-ray transmission spectra comparing the unattenuated baseline (I_0) with beams attenuated by Cu and Zr foils. The distinct Zr K-absorption edge is visible at $\lambda \approx 68.74$ pm.

Figure 3 presents the natural logarithm of the transmission ratio, $\ln(I_x/I_0)$, as a function of wavelength to provide a clearer visualization of the attenuation. A distinctive feature in both representations is observed for Zirconium at a measured wavelength of $\lambda_K \approx 68.74$ pm, where a sharp drop in transmission occurs. This is the K-absorption edge, which occurs when the incident photon

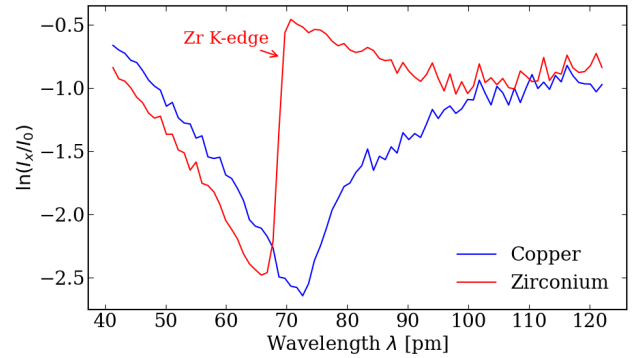


Figure 3: The logarithmic transmission ratio $\ln(I_x/I_0)$ versus wavelength for Cu and Zr. The sharp discontinuity identifies the Zr K-absorption edge.

energy is just sufficient to eject an electron from the innermost K-shell of the Zr atom. This experimental value is in excellent agreement with the theoretical literature value of $\lambda_K = 68.9$ pm [2], yielding a relative error of only 0.23 %.

In contrast, the Copper spectrum does not exhibit an absorption edge within the measured range, as its K-edge is located at $\lambda_K = 138.1$ pm, falling completely outside the experimental window [2]. This confirms that Zirconium ($Z = 40$) has a higher atomic number than Copper ($Z = 29$), resulting in more tightly bound K-shell electrons that require higher energy (shorter wavelengths) to be ejected.

Figure 4 presents a log-log plot of the transmission data to examine the cubic dependence of the absorption cross-section on wavelength ($\tau_a \propto \lambda^3$) outside the ab-

sorption edges. Linear regressions over the $\ln(\lambda)$ intervals of [3.75, 4.15] for Copper and [4.25, 4.50] for Zirconium yielded slopes of 2.51 and 2.42. While these values are somewhat lower than the ideal theoretical exponent of $n = 3$, they are consistent with expected experimental observations for this specific apparatus, where reference data report typical slopes of approximately 2.7 for Copper and 2.5 for Zirconium [6]. This deviation from the ideal cubic relationship arises because the theoretical model primarily accounts for photoelectric absorption, whereas the measured attenuation includes scattering contributions that do not strictly follow the λ^3 power law.

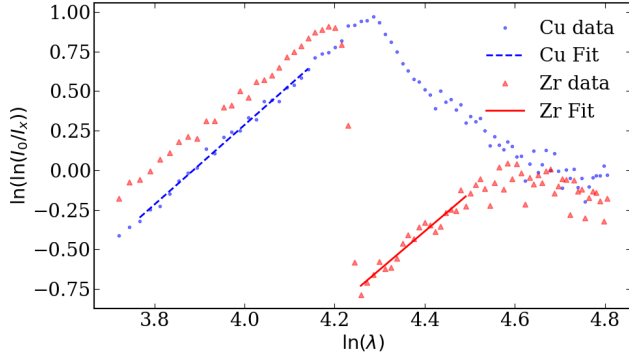


Figure 4: A log-log plot of the absorption parameter $\ln(\ln(I_0/I_x))$ versus $\ln(\lambda)$. The linear fits experimentally verify the λ^3 power-law dependence of X-ray attenuation.

A practical application of these results is the use of Zirconium as a filter for Molybdenum X-ray sources. As demonstrated in Figure 2, the Zr K-edge (68.74 pm) is situated between the Mo K_α (71.67 pm) and K_β (63.85 pm) lines. By utilizing a Zr foil, the K_β line and much of the higher-energy bremsstrahlung are effectively absorbed, while the K_α line passes through relatively unattenuated, resulting in a nearly monochromatic beam [6].

Atomic number dependence

Figure 5 displays the measured counting rates R for five different metal foils (Al, Fe, Cu, Zr, and Ag) as a function of wavelength. The transmission spectra reveal a clear stratification, materials with a higher atomic number exhibit significantly lower counting rates across the measured spectrum, demonstrating a much stronger attenuation of the incident X-ray beam.

To evaluate the atomic number dependence, the atomic absorption cross-section τ_a was calculated for each metal at a fixed wavelength ($\lambda \approx 40.3$ pm) using Eq. (7). Table I summarizes these calculated values alongside the measured transmission data. The data highlight the dramatic impact of atomic number on X-ray transparency.

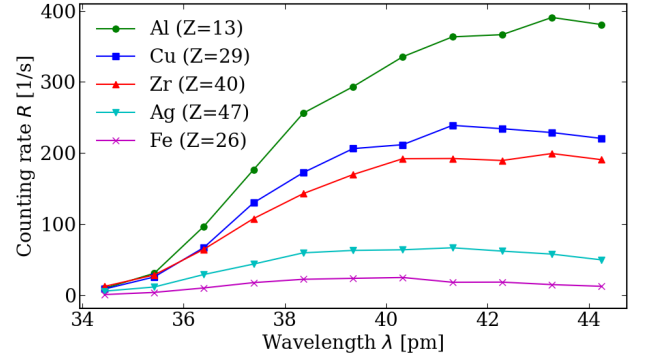


Figure 5: Measured X-ray counting rate R versus wavelength λ for five metal foils. The decreasing transmission with higher atomic number Z demonstrates the strong Z -dependence of attenuation.

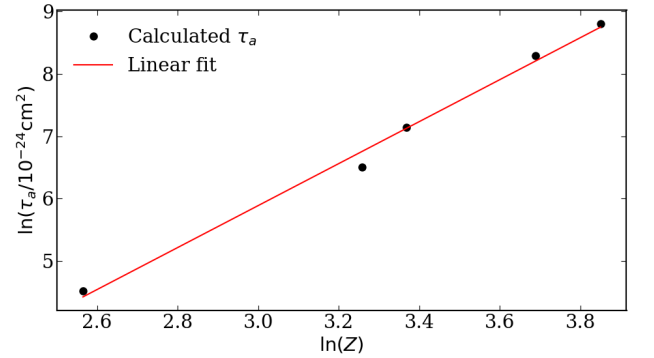


Figure 6: Log-log plot of the atomic absorption cross-section τ_a versus atomic number Z at a constant wavelength.

Aluminum ($Z = 13$) allows 73.9% of the radiation to pass, whereas Silver ($Z = 47$) attenuates 85.9% of the beam.

Table I: Experimental values for the counting rate R , transmission T , and the calculated atomic absorption cross-section τ_a for five metals. The cross-section τ_a is derived from Equation (7).

Metal	Z	R [1/s]	T	τ_a [10^{-24} cm^2]
Al	13	335.4	0.739	91.45
Fe	26	24.9	0.055	666.53
Cu	29	211.7	0.466	1267.94
Zr	40	192.0	0.423	3986.00
Ag	47	63.9	0.141	6653.47

Figure 6 presents a log-log plot of the atomic absorption cross-section τ_a as a function of the atomic number Z to examine the theoretical power-law prediction $\tau_a \propto Z^4$. The linear fit yields a slope of 3.37. While this is somewhat lower than the theoretical integer exponent of $n = 4$, it is a common result in this experimental setup. The deviation from the ideal Z^4 scaling can be attributed to the highly simplified, constant scattering correction (σ_a) and minor uncertainties in the nominal

foil thicknesses [2].

Moseley's law

Figure 7 displays the measured intensity R as a function of wavelength λ for several metallic absorbers (Fe, Mo, Ag, In, and Zr). The primary observation is a systematic shift in the position of the K-absorption edges. As the atomic number Z increases, the edge moves toward shorter wavelengths. For example, Indium ($Z = 49$) exhibits an absorption edge at a much lower wavelength than Zirconium ($Z = 40$). This shift occurs because a higher nuclear charge binds the K-shell electrons more tightly, consequently requiring higher-energy photons (shorter wavelengths) to ionize the atom. It should be noted that for Iron ($Z = 26$), the transmission spectrum remains flat within this experimental window, as its K-edge lies at a much longer wavelength, 174.3 pm [2]. Therefore, the subsequent analysis is restricted to Zr, Mo, Ag, and In.

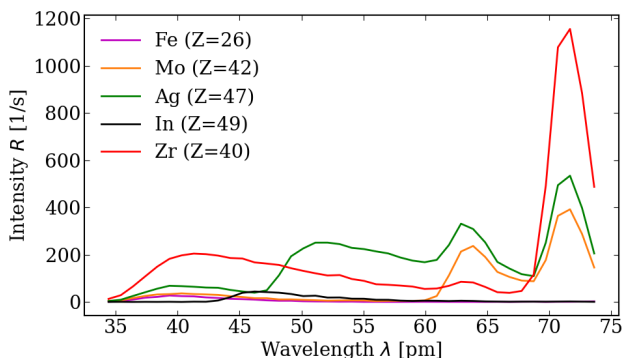


Figure 7: Transmission spectra used to identify the K-absorption edges of various metals. The shift toward shorter wavelengths for higher- Z elements reflects the increased K-shell binding energy.

The K-edge wavelengths obtained from these spectra were converted to binding energies using Eq. (2), and the resulting values were analyzed using the energy form of Moseley's law, Eq. (9). Figure 8 presents the corresponding Moseley plot.

A clear linear dependence is observed in the plotted data, directly supporting the theoretical model. The linear fit yields a slope of $a = 3.70 \sqrt{\text{eV}}$, which corresponds to an experimental Rydberg constant of $R_{\text{calc}} = 1.104 \times 10^7 \text{ m}^{-1}$. This result is in excellent agreement with the theoretical value of $R_{\infty} = 1.097 \times 10^7 \text{ m}^{-1}$ [7], yielding a relative error of approximately 0.6%.

Furthermore, the x -intercept of the linear fit allows for the estimation of the screening constant σ_K . The linear regression yields a value of $\sigma_K \approx 3.71$. While this is slightly higher than the theoretical approximation of $\sigma_K \approx 3.6$ for moderately heavy nuclei [7], the minor de-

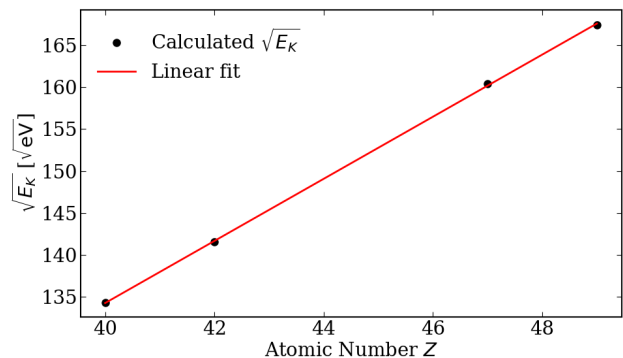


Figure 8: Moseley plot showing the square root of the K-edge absorption energy $\sqrt{E_K}$ versus atomic number Z . The linear relationship experimentally verifies Moseley's law.

viation is physically reasonable. It likely stems from the model underestimating outer-shell shielding, combined with uncertainties in pinpointing the exact K-absorption edges. Nevertheless, this confirms the quantized nature of atomic energy levels and the systematic quadratic dependence of inner-shell binding energies on the effective nuclear charge.

CONCLUSION

In this experiment, the characteristic spectrum of a molybdenum X-ray tube was investigated using Bragg diffraction from a NaCl crystal. The measured spectrum clearly revealed the K_{α} and K_{β} lines of molybdenum and confirmed that Bragg diffraction can be used to determine their wavelengths and photon energies. The fine structure of the K_{α} line could not be resolved in the measured spectrum, indicating that the instrumental resolution of the setup was insufficient. At higher diffraction orders, the angular separation between the $K_{\alpha 1}$ and $K_{\alpha 2}$ lines increases, which improves the possibility of resolving the doublet.

Absorption measurements with metal foils demonstrated the strong dependence of X-ray attenuation on both wavelength and atomic number. A clear K-absorption edge was observed for zirconium, while the copper edge lay outside the measured wavelength range. The results also supported the expected trends that absorption increases with wavelength and with increasing atomic number.

Finally, the measured K-edge energies followed the linear relationship predicted by Moseley's law. The resulting Moseley plot confirmed the systematic dependence of inner-shell binding energies on the atomic number and allowed an experimental determination of the Rydberg constant that was in reasonable agreement with the theoretical value.

-
- [1] J. Als-Nielsen and D. McMorrow, *Elements of Modern X-ray Physics*, 2nd ed. (John Wiley & Sons, Ltd, Chichester, United Kingdom, 2011).
 - [2] LEYBOLD Didactic GmbH, *P6.3.2.3: Investigating the relationship between the coefficient of attenuation and the atomic number Z*, LEYBOLD Physics Leaflets, Germany (n.d.), atomic and Nuclear Physics.
 - [3] LEYBOLD Didactic GmbH, *P6.3.3.1: Bragg reflection: diffraction of x-rays at a monocrystal*, LEYBOLD Physics Leaflets, Germany (n.d.), atomic and Nuclear Physics.
 - [4] R. H. Íslands, *Röntgengeislun (Vinnuseðill)*, University of Iceland (2025), lab manual for Experimental Physics.
 - [5] LEYBOLD Didactic GmbH, *P6.3.3.4: Fine structure of the characteristic x-radiation of a molybdenum anode*, LEYBOLD Physics Leaflets, Germany (n.d.), atomic and Nuclear Physics.
 - [6] LEYBOLD Didactic GmbH, *P6.3.2.2: Investigating the wavelength dependency of the coefficient of attenuation*, LEYBOLD Physics Leaflets, Germany (n.d.), atomic and Nuclear Physics.
 - [7] LEYBOLD Didactic GmbH, *P6.3.3.6: Moseley's law and determination of the Rydberg constant*, LEYBOLD Physics Leaflets, Germany (n.d.), atomic and Nuclear Physics.